

Youngjun Park

Co-founder · CTO, ar-ge inc. | youngjourpark@gmail.com

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Architect and urban data scientist. Co-founder and CTO of ar-ge inc. Ph.D. in Architecture, Seoul National University (2022). Former postdoctoral researcher at KAIST Information and Electronics Research Institute (2022–2025) and KAIST Urban AI Institute (2025). Research in urban computing, smart cities, ML/DL for urban analysis, and Agentic AI for architectural and urban design.

EDUCATION

- **Seoul National University** 2022
Ph.D. in Architecture
 - *Dissertation:* Analysis of pedestrian activity using smartphone data, and urban planning facilities for the promotion of physical activity
- **Seoul National University** 2017
M.S. in Urban Design
 - *Thesis:* A study on homemakers' automobile use behavior and intra-neighborhood travel characteristics
- **Seoul National University** 2013
B.S. in Architecture

EXPERIENCE

- **ar-ge Architects (ar-ge inc.)** 2025 – Present
Co-founder · CTO
 - Development of Agentic AI-based architectural and urban design tools
 - Leading the research and engineering organization
 - Public-sector architectural and urban design practice
- **KAIST Urban AI Institute** 2025
Postdoctoral Researcher
 - Supported institute operations
 - Managed interdisciplinary and international collaborative research projects
 - Research on multi-agent systems and Urban Foundation Models
 - Participated in AI Seoul 2025 and exhibited MIT–KAIST research outcomes
- **KAIST Information and Electronics Research Institute** 2022 – 2025
Postdoctoral Researcher
 - Research on smart city technologies based on IoT edge computing
 - Urban data processing methodologies including spatiotemporal analysis, graph networks, and LLM embeddings
 - Led 'DeepUrban', the urban research group within the lab
- **Department of Architecture, Seoul National University** 2021 – 2024
Lecturer
 - Studies in Urban Design: Pedestrian–Urban Data Analysis (graduate course, Architecture)
 - Theories of Urban Design: Traditional and Contemporary Urban Design (graduate course, Interdisciplinary Program in Urban Design)
- **Republic of Korea Army Corps of Engineers** 2013 – 2015
Officer (First Lieutenant)
 - Led live operational missions and unit management
 - Mine-detection-and-removal team leader for DMZ operations

AWARDS AND HONORS

- **G-School Best Innovation Award** 2023
KAIST–NYU Young Researcher Day
- **First Prize, AI-based Population Prediction** 2023
COMPAS Data Analysis Competition, Korea Land & Housing Corporation (LH)
- **Outstanding Graduation Thesis Award** 2018
Master's thesis, Architectural Institute of Korea
- **Best Prize** 2017
Design Works Exhibition, Graduate School of Environmental Studies, Seoul National University
- **Selected as Future Architectural Education Talent** 2016
Korean Institute of Architects

RESEARCH AND PUBLICATIONS

- [J.6] **Park, Youngjun** and Lee, Sejin (2026). **Virtual Public Hearing Simulation using Large Language Model-based Multi-Agent System: Focused on the Public Hearing for the Rural Spatial Restructuring and Regeneration Plan.** *Journal of the Urban Design Institute of Korea*, 27(4), 61–78.
- [J.5] **Park, Youngjun**, Na, Jihye, Jang, Keonhee, Lee, Dongman, Kwon, Changhyun, and Yoon, Yoonjin (2026). **Food Deserts in Context: How Site-Specific Factors Reshape Food Deserts Discussion.** *npj Urban Sustainability*.
- [C.6] Lee, Hongju, **Park, Youngjun**, An, Jisun, and Lee, Dongman (2025). **Enhancing Regional Airbnb Trend Forecasting Using LLM-Based Embeddings of Accessibility and Human Mobility.** *ASONAM 2025*.
- [C.5] **Park, Youngjun**, Bae, Hyungchan, An, Jisun, and Lee, Dongman (2025). **A Model for Monthly, Local-Level Airbnb Changes Using Public Dataset.** *Proceedings of the 7th International Conference on Tourism Research*, 8(1), 214–223.
- [Po.1] **Park, Youngjun** and Han, Sumin (2024). **Enhancing Population Predictions in Developing Area through Building Construction Data.** *Association of European Schools of Planning (AESOP'24)*, poster session.
- [W.1] Han, Sumin, **Park, Youngjun**, Sabir, Sonia, An, Jisun, and Lee, Dongman (2024). **Improving Real Estate Appraisal with POI Integration and Areal Embedding.** *DMO-FinTech'24: Workshop on Decision Making and Optimization in Financial Technologies (PAKDD'24)*.
- [C.4] **Park, Youngjun**, Han, Sumin, Park, Doyun, Bae, Hyeongchan, Nguyen, Thanh Tung, and Lee, Dongman (2024). **Integrative Geospatial Visualization for Urban Mobility Analysis.** *Korea Computer Congress (KCC'24)*.
- [C.3] Elmoursi, Ahmad, Han, Sumin, Bae, Hyungchan, **Park, Youngjun**, Nguyen, Thanh Tung, and Lee, Dongman (2024). **Enhancing Geospatial Recommendations through Natural Language Processing with Large Language Models.** *Korea Computer Congress (KCC'24)*.
- [C.2] Han, Sumin, **Park, Youngjun**, Lee, Minji, An, Jisun, and Lee, Dongman (2023). **Enhancing Spatio-temporal Traffic Prediction through Urban Human Activity Analysis.** *CIKM 2023*, 689–698.
- [C.1] **Park, Youngjun**, Han, Sumin, An, Jisun, and Lee, Dongman (2023). **Encoding Urban Trajectory as a Language: Deep Learning Insights for Human Mobility Pattern.** *Association of Collegiate Schools of Planning (ACSP '23)*.
- [J.4] **Park, Youngjun**, Lee, Sunjae, and Park, Sohyun (2022). **Differences in park walking, comparing the physically inactive and active groups: data from mHealth monitoring system in Seoul.** *International Journal of Environmental Research and Public Health*, 19(1), 395.
- [O.2] **Park, Youngjun** (2022). **Parks and Healthy Walking: smartphone physical activity recognition data for place and walking.** *Urban Design Institute of Korea*.
- [J.3] **Park, Youngjun**, Chung, Haegwon, and Park, Sohyun (2021). **Changes of walking activity during the first cycle phases of COVID-19 pandemic: A case study of Seoul, Korea.** *HERD: Health Environments Research & Design Journal*, 14(4), 43–57.
- [J.2] Jeong, Jiwoon, **Park, Youngjun**, and Park, Sohyun (2021). **Safety-critical events in bicycle lanes in Jongno, Seoul.** *Journal of Urbanism: International Research on Placemaking and Urban Sustainability*, 16(1), 1–24.
- [J.1] **Park, Youngjun** and Park, Sohyun (2019). **Car-use and Non-work Travel of Housewives in Residential Neighborhood of Seoul.** *Seoul Research*, 20(1), 45–59.
- [O.1] **Park, Youngjun** and Park, Sohyun (2017). **A Study on the Location of Car-sharing in the Neighborhood Area and Replacement of Vehicle Ownership in Community.** *Urban Design Institute of Korea*.

RESEARCH PROJECTS AND GRANTS

- **GS-KAIST Smart City Research Center** 2024–2025
Postdoctoral Researcher · KAIST · GS E&C
- **AI-based Non-contact Drug Detection Technology** 2024–2025
Postdoctoral Researcher · KAIST · Ministry of Science and ICT · RS-2024-00459749
- **Multimodal Spatiotemporal Urban Prediction Models Based on Large Language Models** 2024–2025
Postdoctoral Researcher · KAIST · National Research Foundation of Korea · RS-2024-00356597
- **Self-Learning Autonomic IoT Edge Computing Technology** 2022–2025
Postdoctoral Researcher · SW StarLab Project, KAIST · Ministry of Science and ICT · RS-2019-II191126
- **Development of an AI-based Smart Mobility System** 2022–2024
Participating Researcher · Daegu Metropolitan City · KAIST · N04230030
- **Text Analysis of Pedestrian-Related Government-Budget Projects** 2022–2023
Research Service · Architecture & Urban Research Institute (AURI)
- **Comparative Study on Walking Practice within Neighborhoods via Individual GPS Trajectory Analysis** 2019–2021
Research Grant Recipient · National Research Foundation of Korea, Next-Generation Researcher Program · Seoul National University

- **Pedestrian Data Analysis for WalkON Smartphone App Development** 2019–2020
Research Team Lead, Affiliated Researcher (External Organization) · Swallaby Inc.
- **GIS Analysis for Spatial Welfare of Infants and Young Children** 2018–2019
Research Assistant, Affiliated Researcher (External Organization) · Dept. of Architecture, Ewha Womans University
- **Comprehensive Urban Plan for the Huế City ODA Project, Vietnam** 2015–2016
Research Assistant · Korea International Cooperation Agency (KOICA) · Graduate School of Environmental Studies, Seoul National University

PATENTS AND INTELLECTUAL PROPERTY

- **Method and apparatus for estimating housing value using a neural network model** 10-2024-0108083
Patent (Filed) KR
- **Method and server for providing commercial real estate information** 10-23679770000
Patent (Registered) KR
- **Geographic data visualization program** C-2023-062275
Copyright (Registered) KR

PROFESSIONAL SERVICE

- **Journal Reviewer** Scientific Reports · BMC Public Health · Archives of Public Health · Frontiers in Public Health
- **Research Advisory** KAIST Urban AI Institute · Architecture & Urban Research Institute (AURI)
- **Industry–Academic Collaborations** GS E&C · Kia Motors–Innocean · Swallaby
- **Invited Lectures · Seminars** Dept. of Architecture, Seoul National University · KAIST School of Computing · Dept. of Architectural Engineering, KOREATECH · Dept. of Architecture, Chung-Ang University

TEACHING

- **Urban Design Research — Urban Walking** 2020–2021
Graduate course, Architecture major · Seoul National University, Dept. of Architecture
- **Theories of Urban Design: Traditional and Contemporary** 2020–2021
Graduate course · SNU Interdisciplinary Program in Urban Design